

Lecture 5 - Consumer and Producer Surplus

ECON 101
Fall 2020



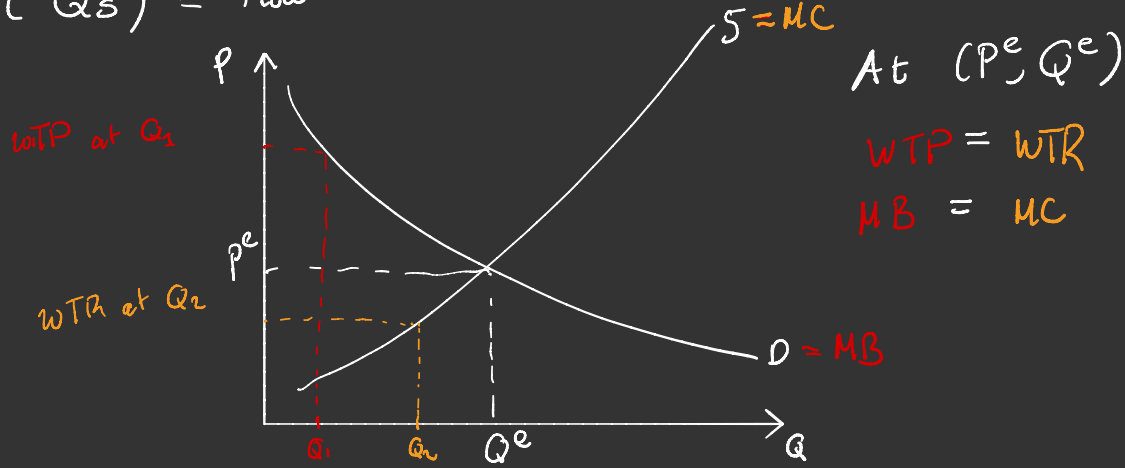
An alternative interpretation of Demand and Supply

- ① $Q_D = Q_D(P, DET^D)$ } How much is each participant
② $Q_S = Q_S(P, DET^S)$ } willing to demand and
offer at each price,

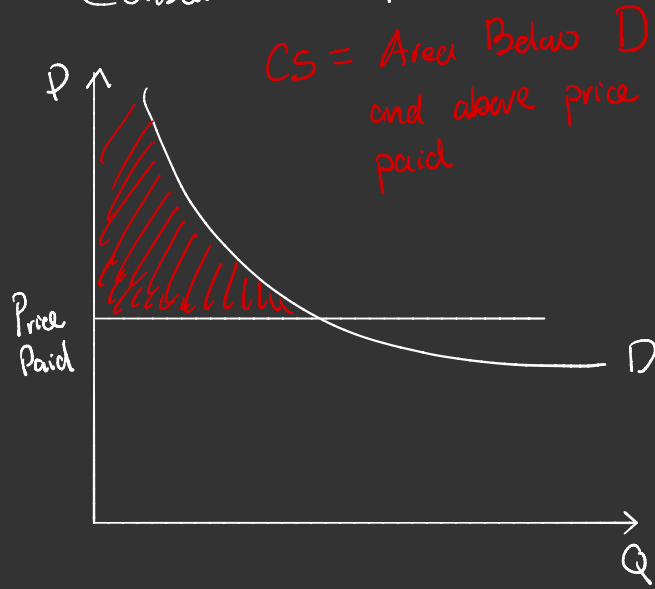


Inverse schedules; for a fixed fixed level of the determinants:

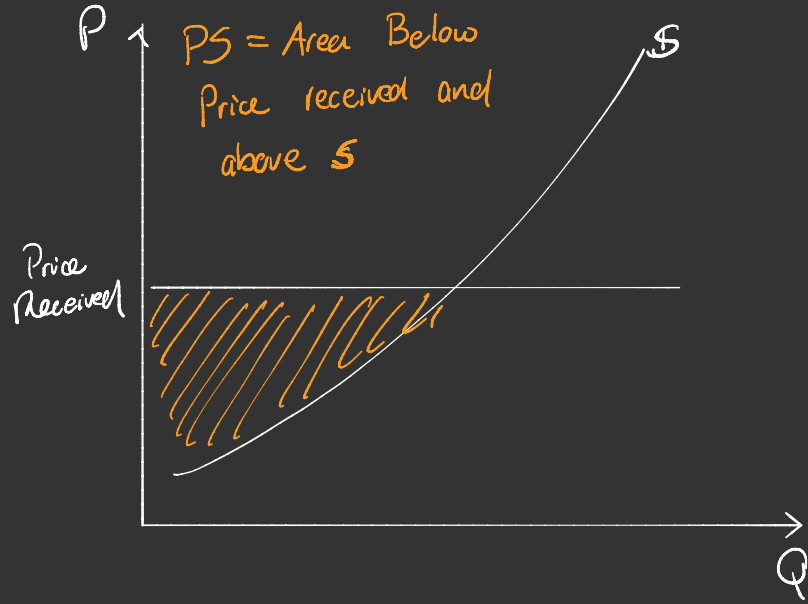
- ① $P_D = P_D(Q_D) =$ How much the consumer is willing to pay: $WTP = \text{Marginal Benefit}$
② $P_S = P_S(Q_S) =$ How much the seller is willing to receive: $WTR = \text{Marginal Cost}$



Consumer Surplus



Producer surplus



$$TS (\text{Social Surplus}) = CS + PS$$

For Supply and demand: CS and PS are difficult to calculate

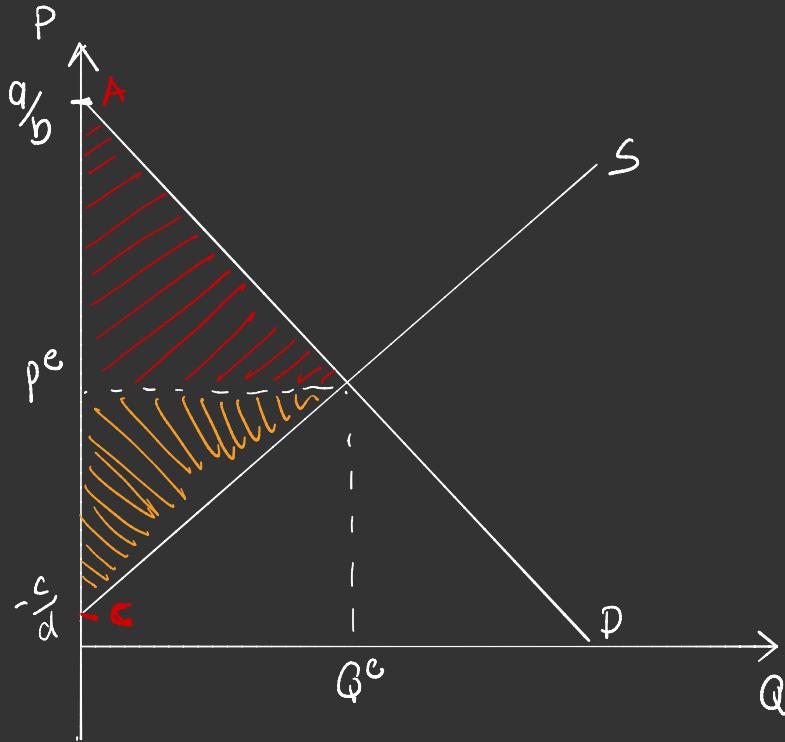
Linear demand & supply schedules

$$Q_D = a - bP \quad a, b, c, d > 0$$

$$Q_S = c + dP \quad a, b \text{ are placeholders for the determinants}$$

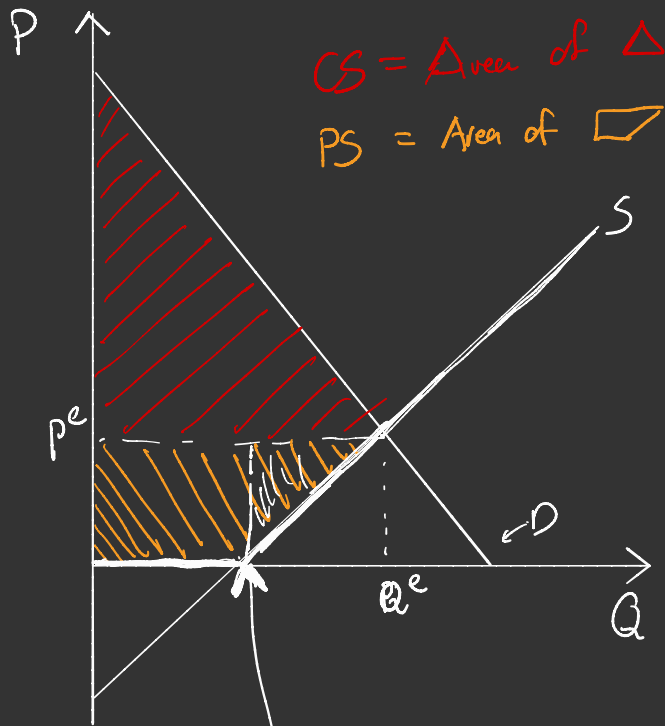
$$P^e = \frac{a-c}{b+d}$$

$$Q^e = \frac{ad-bc}{b+d}$$



$$CS = \text{Area of } \triangle AP^eQ^e = \frac{(a/b - P^e) Q^e}{2}$$

$$PS = \text{Area of } \triangle P^eQ^eC = \frac{[P^e - (c/d)] Q^e}{2}$$



CS = Area of $\triangle$

PS = Area of $\square$

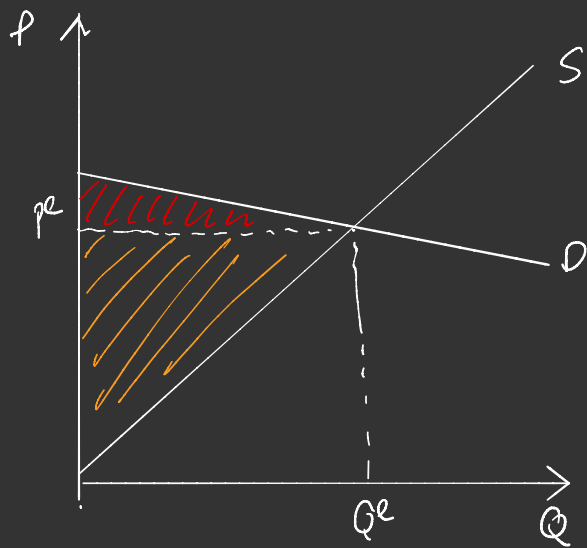
← To compute this you need to know the horizontal intercept of the inverse supply curve

$$Q_S = c + dP \quad c, d > 0$$

$$\text{Producer Surplus: } P^e \times Q_S^0 + \frac{(Q^e - Q_S^0) P^e}{2}$$

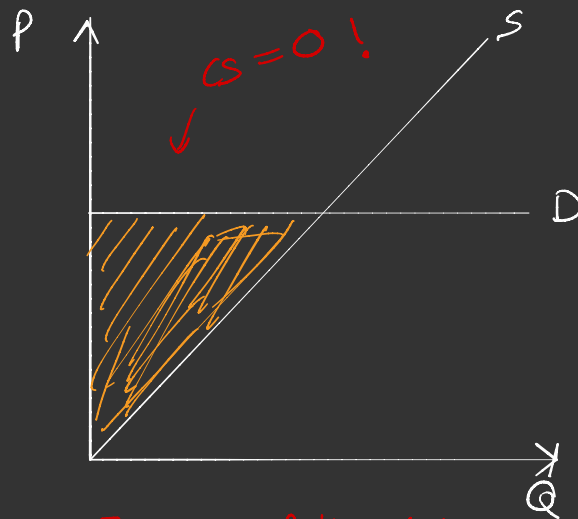
c is the horizontal intercept, call it $Q_S^0 = c$

These sellers are willing to part with this quantity at a price of zero: c is the horizontal intercept!

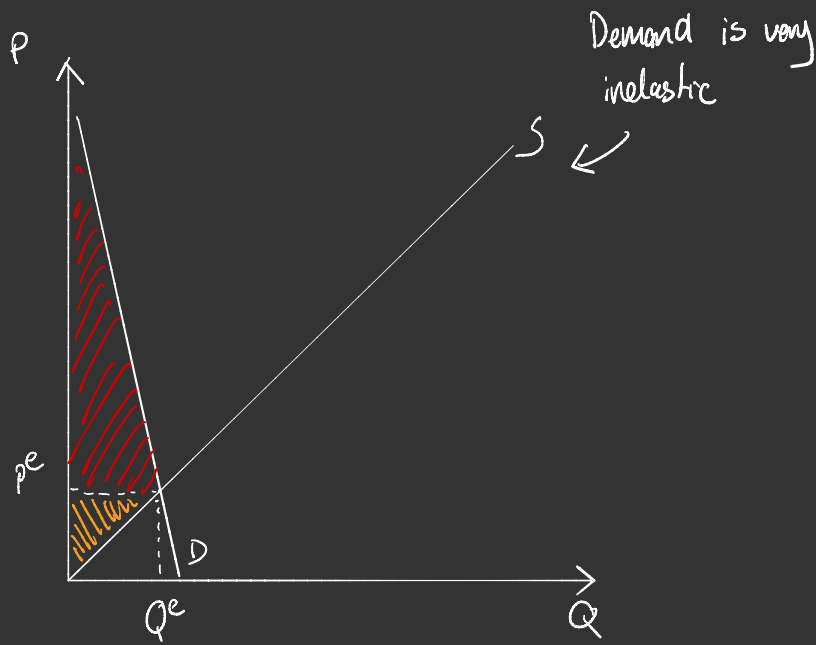


$$CS + PS = TS$$

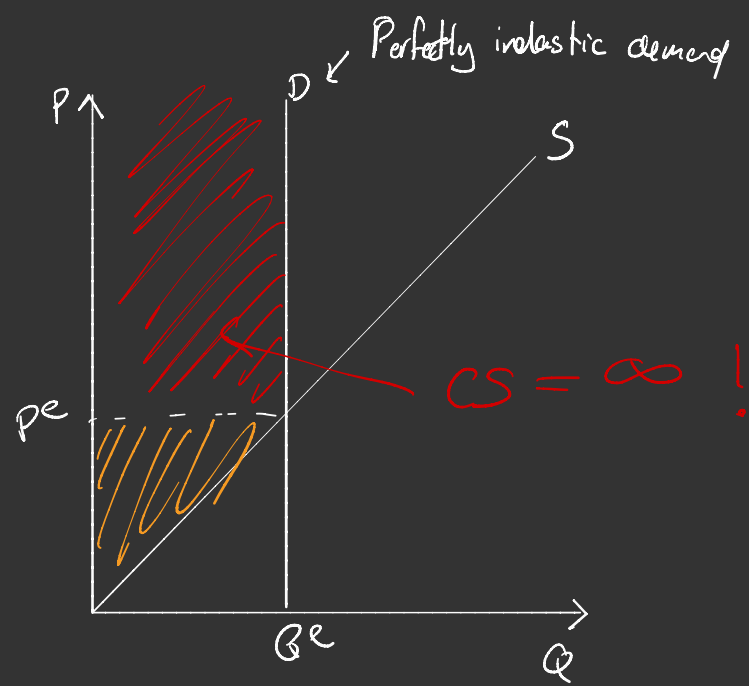
Demand is very flat $\Rightarrow$ very elastic



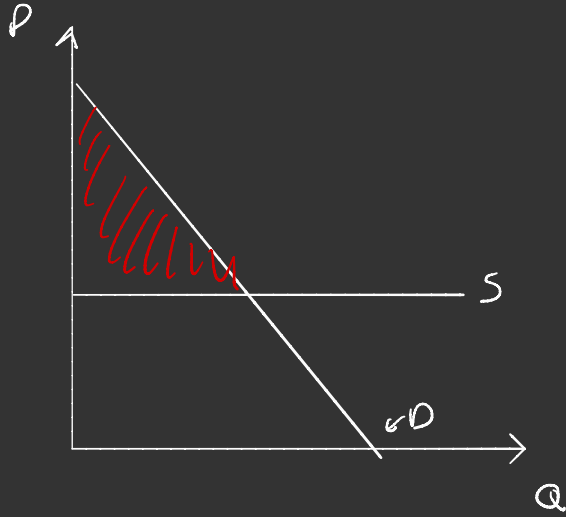
For a perfectly elastic demand $CS=0$



$$TS = CS + PS$$

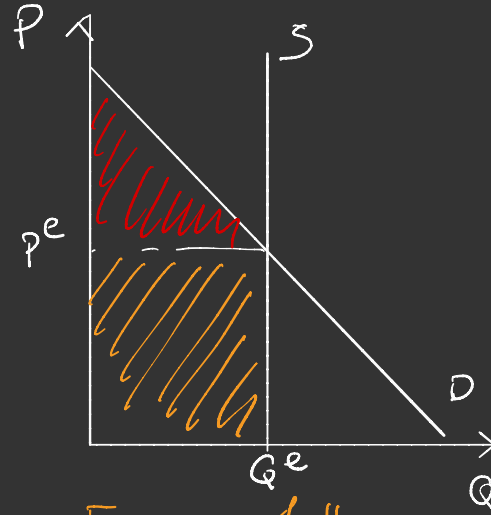


CS is ∞ for a perfectly inelastic demand

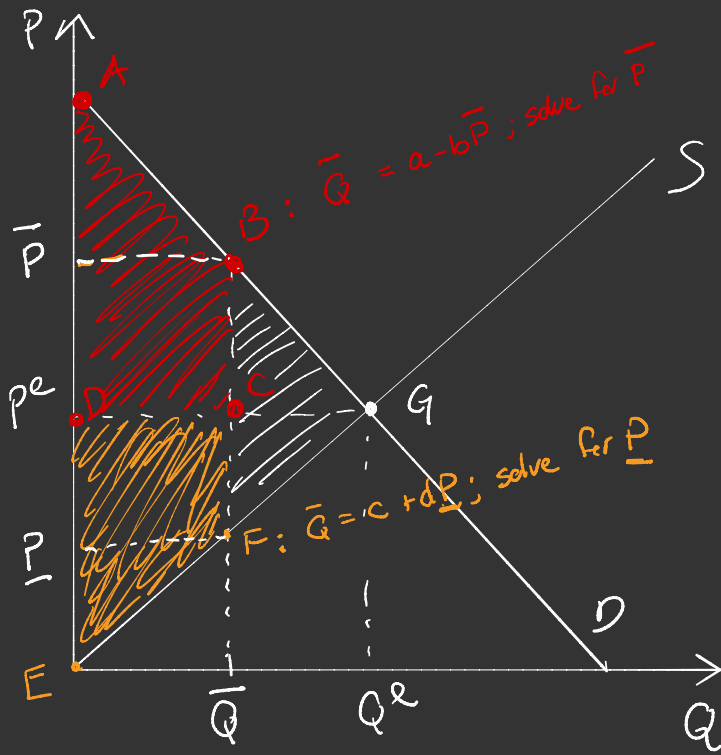


For a perfectly elastic supply,
 $PS = 0!$

$$TS = CS + PS$$



For a perfectly inelastic supply
 Note that $PS = P^e \times Q^e = \text{Revenue}$



Deadweight loss

Suppose Quantity is restricted
at $\bar{Q} < Q^e$

CS = Area of ABCD

PS = Area of DCEF

DWL = Area BFG